Package

To organize the related files

Access modifiers – class – public and default

d.m – public, private, protected and default

can we use public, private, protected and default to local variables ?

error , local variables are accessible only in that block.

Java is not allowing to use access modifier to the local variable.

static k.w to the local variable - error

scope of the static keyword should be through out the class.

Package

Public class in a package

Same package, from other package using import

Default class in package

Inside the same package, from other package even you add import stmt won't work.

Public class with public d.m In a package

Inside the same class, from other class in the same package

From other package using import

Public class with private dm

Only with in the class

Public class with default dm (private with in a package)

Inside the same class, other classes in the same package

Public class with protected dm

Inside the same classes

Other classes in the same package without inheritance

From other package using inheritance

Decision making stmt

If, if else, nested if, if else if ladder , switch case

== and equals

String name1 = “abi”;

String name2 = “abi”;

name1.equals(name2); checks the content of the string object

name1==name2; == checks the references

compareTo method? ASCII

nextLine () reads the enter key as a input

next() method removes that enter key input from the keyboard buffer. Because of this , need not to add a stmt

scanner.nextLine(); to take enter key as a input.

Next method removes the enter key from the input buffer. But it skips reading the string after the space.

Difference between while loop and do while loop

If the condition is true then only logic inside the while loop will be executed.

Even though the condition is false, executes the logic one time. Why because in do while we check the condition at last.

Array

In array we can store n number of values of same type and memory is allocated in continuous manner.

Employee employee = new Employee();

Employee employee; // declaration

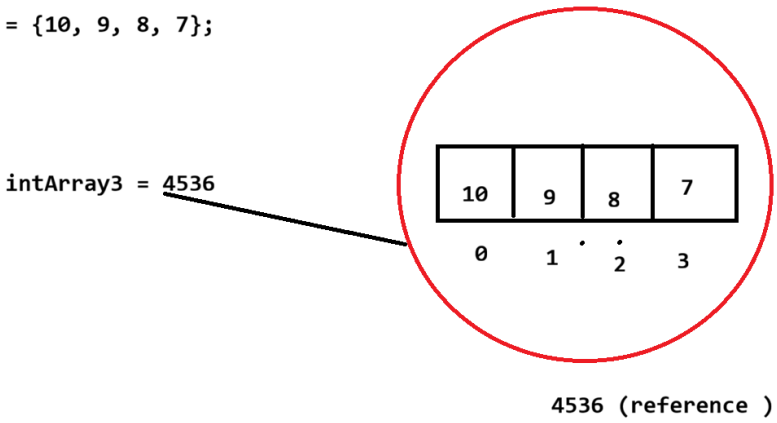
employee = new Employee(); //instantiation

```
int intArray2[];//array declaration
intArray2 = new int[4]; // array instantiation
```

in java array is a object. Array reference variable holds the reference of the created object.

Single dimensional array

```
int intArray3[] = {10, 9, 8, 7};
```



intArray[2]

intArray[2] = 200;

to print the single dimensional array values.

```
Arrays.toString(intArray3)
```

Arrayvariable.length gives the length of the single dim array

```
for (int i : array)
        {
            System.out.println(i);
```

}

for(variable to hold the single element from the array : array variable)

{

}

A 2D array is an array that contains other arrays.

When we can use single dim array?

To store list of values 1, 2, 3, 4, 5

When we can use 2 dim/multi dim array?

To store the table structure values

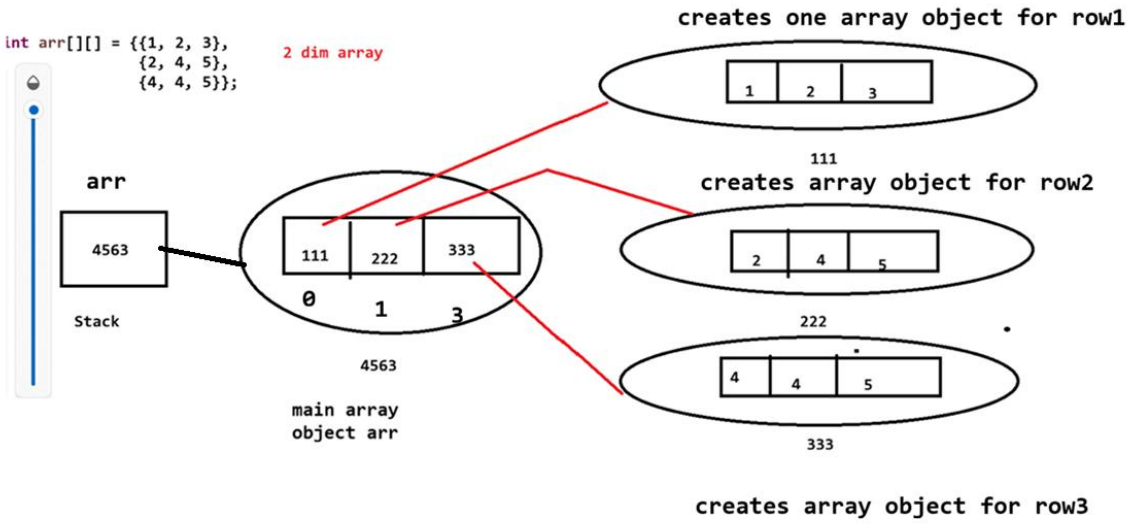
1 2 3

4 5 6

7 8 9

arr.length = 3 rows

arr[0].length = 3 columns



```
Arrays.deepToString(tArray)
```

To print the multidim array without using for /for each loop

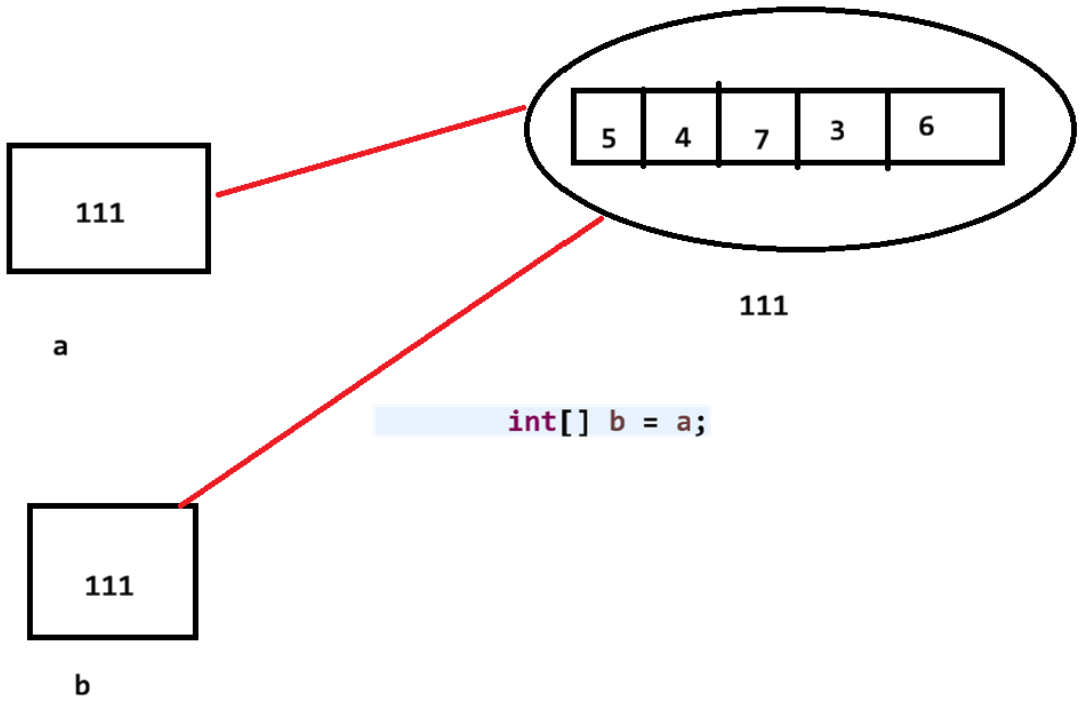
a =10;

b = a;

int age = 30;

Employee employee; employee ref variable we can store the reference of type Employee

```
int[] j; array object of type int
```



Since both reference variable a and b holding 111, making change in one array affects another array.